

Jyoti Prakash Patra

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SUMMARY

Senior Backend and Platform Engineer with 7+ years of experience building distributed, event-driven systems and LLM-powered platforms at production scale. At Maersk, architected an AI documentation assistant (DevPal) serving 18+ engineering teams using hybrid RAG, Azure OpenAI, and MCP — and independently owns a high-traffic survey platform built on Kafka, Temporal, and Node.js. Deep experience across the full delivery lifecycle: infra provisioning, schema design, CI/CD, LLM observability, and production migrations.

EXPERIENCE

Senior Software Engineer

Maersk

Nov 2023 – Present

Bangalore, India

InsightsHub — Enterprise Survey & Feedback Platform

- Led end-to-end migration of a production survey platform from Java Spring Boot to **Node.js (Fastify)**, reducing cold-start latency by ~45% and cutting infrastructure costs by ~30%.
- Provisioned **Azure PostgreSQL Flexible Server** via Terraform; owned CI/CD pipelines from build to production deployment.
- Redesigned data schemas with **Prisma ORM** and executed zero-downtime large-scale production data migration with automated validation and rollback safeguards.
- Integrated **Kafka** consumers to synchronize application and collaborator metadata with central systems, processing 10K+ events per day reliably.
- Built **Temporal** workflows for survey scoring (NPS, CSAT, SUS), notifications, and collaboration updates — replacing fragile cron jobs and reducing failure-related incidents by ~70%.
- Implemented **Redis** caching layer supporting 1,000+ concurrent users with sub-50ms read latency, preventing DB saturation under peak load.
- Created a dedicated orchestration service running Temporal workers, enabling horizontal scaling of async workloads independently from the API layer.
- Set up **Prometheus + Grafana** dashboards tracking latency percentiles, error rates, and throughput; reduced mean time to detect (MTTD) issues by ~50%.

Feature Suggestions Platform — Internal Product Feedback System

- Designed and delivered a cross-team feature suggestion system serving multiple product teams, enabling structured product feedback at scale.
- Modeled hierarchical discussions using **self-referential Prisma relations**, supporting arbitrarily nested comments and replies with $O(1)$ lookup per node.
- Implemented soft-delete strategies to preserve thread integrity and auditability across comment lifecycles.
- Built engagement features (likes, emoji reactions, voting) and email notification workflows for comments, replies, and feature status transitions.
- Designed a feature lifecycle integrated with **Jira**, enabling cross-team triage, scheduling, and release tracking from a single interface.

DevPal (DevEx Agent) — Enterprise LLM Developer Platform, 18+ Teams

- Architected a multi-tenant AI documentation assistant in **FastAPI + Azure OpenAI** serving 18+ Maersk engineering teams, indexing 1,640+ documents and 1,107 GitHub discussions with 3072-dimension embeddings.
- Designed a **hybrid RAG pipeline**: 3-pass LLM shuffle-and-tag ingestion strategy → Reciprocal Rank Fusion (RRF) merging tag-based search, vector similarity, and PostgreSQL full-text search — with LLM re-ranking via **GPT-4.1-mini** before answer generation via **GPT-4.1**.
- Eliminated OpenAI Agents SDK overhead by replacing the gpt-4.1 routing agent with a direct **gpt-4.1-mini structured output call** — reducing P50 response time from 33–40s to 22–27s (~35% faster) and streaming first token from 37s to 2–5s (10× improvement).
- Reduced GitHub Discussions initial embedding sync from **55 minutes to ~5 minutes** by replacing sequential per-item API calls with batched embedding generation (100 texts/request).
- Designed **AI Ticket Triage service** (TriageService): LLM completeness check → hybrid retrieval → confidence-gated auto-reply (threshold: 0.70) → Hedwig on-call escalation; integrated with MS Teams via Power Automate and GitHub Discussions webhooks; built bot-loop prevention via triage state machine.

- Built a **platform tool registry** auto-generated from OpenAPI specs (MDP, MTTS, MRMP, MAC, Stargate APIs) enabling the LLM to fetch live platform data — activity logs, service state, Kafka topics, team rosters — without leaving the chat interface; user bearer token forwarded, no service principal needed.
- Exposed an **MCP server** with Azure AD OAuth, enabling DevPal to be consumed as a tool in VS Code, Microsoft Teams, and Jira — adopted by 100+ engineers across the organization.
- Built a **35-panel Grafana + Prometheus** production observability stack; diagnosed and fixed 2 silent instrumentation bugs (fallback counter never incrementing, latency histogram using wrong label) that had been masking real metrics in prod.

Software Engineer

Jul 2022 – Oct 2023

HealthEdge Technologies

Bangalore, India

- Built a reusable **React + Redux** component library published as internal NPM packages, reducing UI development time across 3 product teams by ~25%.
- Developed backend REST APIs in **Java Spring Boot** with JWT-based authentication and role-based access control (RBAC).
- Implemented BPMN-based process workflows using **Kogito** and managed schema versioning via Liquibase across multiple environments.

Software Developer

Feb 2019 – May 2022

Tata Consultancy Services

Hyderabad, India

- Developed and maintained REST APIs for enterprise SaaS applications serving 10,000+ users using **Java** and **Spring Boot**.
- Wrote unit and integration tests achieving >80% code coverage on critical service modules.
- Automated operational workflows using shell scripts, reducing manual release effort by ~3 hours per deployment cycle.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Backend: Node.js, Fastify, Express.js, FastAPI, Spring Boot, REST APIs, Microservices, SSE Streaming

Distributed Systems: Apache Kafka, Temporal, Apache ActiveMQ, Apache Camel, Event-Driven Architecture

Databases: PostgreSQL (tsvector, pgvector), Prisma ORM, Redis, Schema Design, Zero-Downtime Migrations

Cloud & Infra: Azure, Azure OpenAI, Azure AD / OAuth 2.0, Terraform, Docker, Kubernetes, CI/CD

Observability: Prometheus, Grafana, Arize Phoenix (LLM Tracing), OpenTelemetry, SLO/SLA Monitoring

AI / LLM: Hybrid RAG, RRF Retrieval, Vector Embeddings, Semantic Tagging, MCP (Model Context Protocol), LLM Tool Use, Structured Output, Evals (Precision/Recall/F1)

Testing: pytest, Jest, Integration Testing, TDD, LLM-as-Judge Evaluation

EDUCATION

Parala Maharaja Government Engineering College

Bachelor of Technology, Mechanical Engineering

Odisha, India

Aug 2014 – May 2018